Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/24/2017
PI Name:	Chahine, Georges D.Sc.		
Project Title:	Two-Phase Flow Separation and Characterization for Microgravity Applications		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Jessup	**State:**	MD
Zip Code:	20794	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/21/2015	**End Date:**	10/20/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:			
Flight Assignment:	NOTE change in grant number to 80NSSC18M0001 per NSSC information and J. McQuillen/GRC (Ed., 1/16/2018)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Xiongjun (Dynaflow, Inc.) Loraine, Greg Ph.D. (Dynaflow, Inc.)		
Grant/Contract No.:	80NSSC18M0001 ; NNX16AB85A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator," grant NNX11AO76A with new grant NNX16AB85A and title, "Two-Phase Flow Separation and Characterization for Microgravity Applications" and same Principal Investigator Georges Chahine, D.Sc. (Ed., 4/18/2016); 1/16/2018 update: New grant number 80NSSC18M0001 awarded for same title--Two-Phase Flow Separation and Characterization for Microgravity Applications (Ed., 1/16/18) The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse. On Earth, bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second, the tank is fixed and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50-80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of a separator for low void fractions. We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirling flow inside the inner cylinder is generated with the help of tangential injection slots. With this configuration one can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guarantees removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. These are then removed through two central exit orifices at each end of the cylinder. This configuration was tested successfully during a reduced-gravity flight and we are planning additional testing. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in earth applications where fast separation of low void fraction gas is of concern.
Task Progress:	During this reporting period, we have conducted ground tests and computational fluid dynamics (CFD) simulations on the DYNASWIRL® phase separator and worked with the NASA engineering team supported by Zin Technologies, to help develop documents for the Requirements Definition Review (RDR). We have investigated the effects on the separator of changes in the requirements requested by the engineering team in order to be able to fit the DYNASWIRL® phase separator in the test rig to deploy onboard International Space Station (ISS). Progress made concerned the following tasks: A. High Speed Imaging Requirements: Understanding the flow field and bubble dynamics inside the phase separator was an important task we worked on in order to define high-speed imaging requirements. Extensive PIV (Particle Image Velocimetry) measurements were conducted in the laboratory at DYNAFLOW to obtain the velocities in the swirl flow field under different flow rate conditions and at different axial locations. We also conducted extensive experiments to resolve the bubble dynamics for bubbles of various sizes and used this to identify the required frame rate, exposure time, and required lighting. These experiments also provided us data on the ground that can be used in the future to compare with the data to be obtained under reduced gravity on the ISS. B. Two-Phase Flow Numerical Simulations: 3D Numerical simulations were conducted by coupling our 3DYNAFS-VIS©, a two-phase flow Navier-Stokes solver to model the overall flow field of the bubbly mixture in the DYNASWIRL® phase separator, and 3DYNAFS-DSM©, a bubble tracking and bubble dynamics motion solver. Both components are modules of our 3DYNAFS © CFD suite and are coupled at each time step. The results from the numerical simulation were used to investigate the effects of bubble size, flow conditions, and gravitational conditions on the gas collection into the core and its extraction. These data helped us refine the SRD (Science Requirement Document) and test matrix. C. SRD Updates: Besides experiments conducted to assist the definition of the high-speed imaging requirements, we have also conducted experiments to fine tune and update the SRD, including identifying good locations to place pressure ports, scheme to fabricate the gas extraction membrane tube, etc. We have been actively working with the team members from NASA Glenn Research Center (GRC) and Zin Technologies to refine the SRD through several iterations. D. Test Matrix Updates: Due to the constraints imposed by the available resources on the ISS (space, power, data storage, electromagnetic field, …etc.), the test matrix has been under regular updates during this reporting period. Refinements through multiple iterations in order to maximize the coverage of the scientific objectives within the limits of the available resources. Both ground experiments and numerical simulations were conducted based on the imposed constraints to provide data to optimize the test matrix. E. Instrumentation Development for Void Fraction Measurement: We continued our efforts to develop an instrument based on electrical impedance measurements to obtain the void fraction of mixture at high void fractions. We have managed to remove signal drift by monitoring the temperature change of the liquid, which can eliminate the requirement of a reference probe to compensate for the signal drift. Tests conducted at different liquid and gas flow rates indicate that, in our test loop, the new scheme without a reference probe can give good measurements of void fractions up to 50%.
Bibliography Type:	Description: (Last Updated: 09/04/2025)

Articles in Peer-reviewed Journals	Hsiao C-T, Ma J, Chahine GL. "Numerical study of gravity effects on phase separation in a swirl chamber." Chemical Engineering Science. 2017 Jun 29;165:177-85. https://doi.org/10.1016/j.ces.2017.03.005 , Jun-2017
Articles in Peer-reviewed Journals	Wu X, Loraine G, Hsiao C-T, Chahine GL. "Development of a passive phase separator for space and earth applications." Separation and Purification Technology. 2017 Dec;189:229-37. (Available online 7 August 2017.) https://doi.org/10.1016/j.seppur.2017.08.017 , Dec-2017